

Regression for Binary Data

June 2025

Data Source

Original source: <http://datadryad.org/resource/doi:10.5061/dryad.sq72d>

Our version at: <https://sldr.netlify.app/data/FrogAbnormalities.csv>

Living Dead Frogs | World's Weirdest

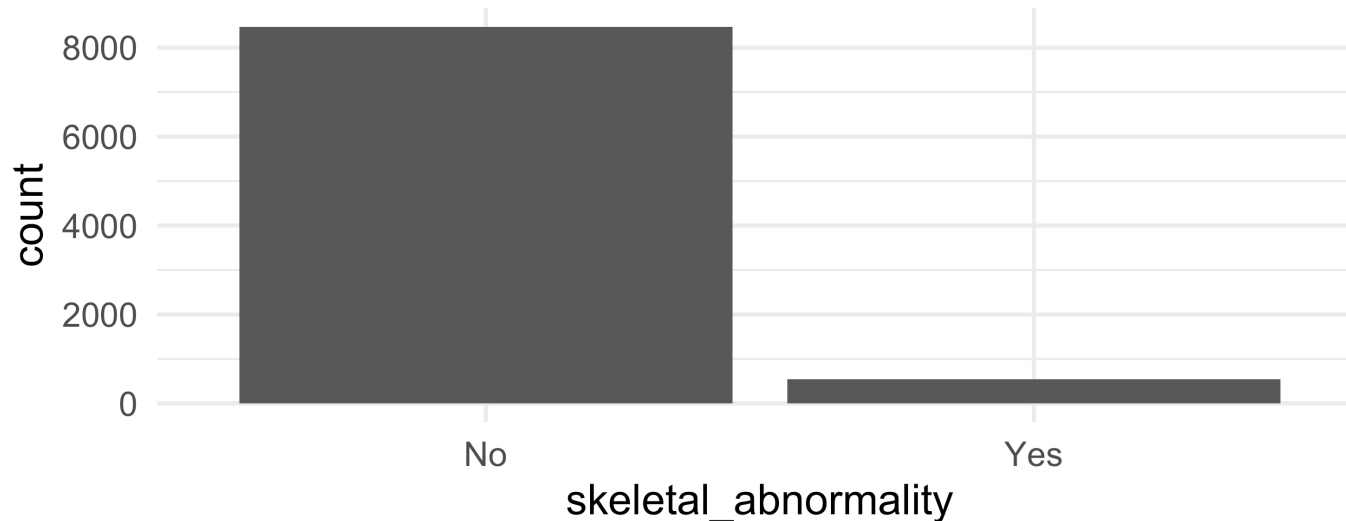


Variables in the dataset include:

- IDs: `collection_id`, `frog_id`, `site`
- Info on time of data collection: `date`, `year`, `coll_date`
- Size and developmental stage of the frog: `gosner_stage`, `tail_length` (which is longer for young frogs, that is, tadpoles)
- Whether or not the frog has any abnormality in general (`abnormal`), an injury (`bleeding_injury`), or a specific type of abnormality: `skeletal_abnormality`, `eye_abnormality`, `surface_abnormality`
- Relative abundance of invertebrate predators of frogs: `dragonfly_relative_density` and `other_invert_relative_density`
- *Rough* water testing results: `detectable_analytes` (average number of contaminants present)

Binary Response

- `abnormal` has values "no" and "yes"
- which do we want to call *success* & predict probability of?



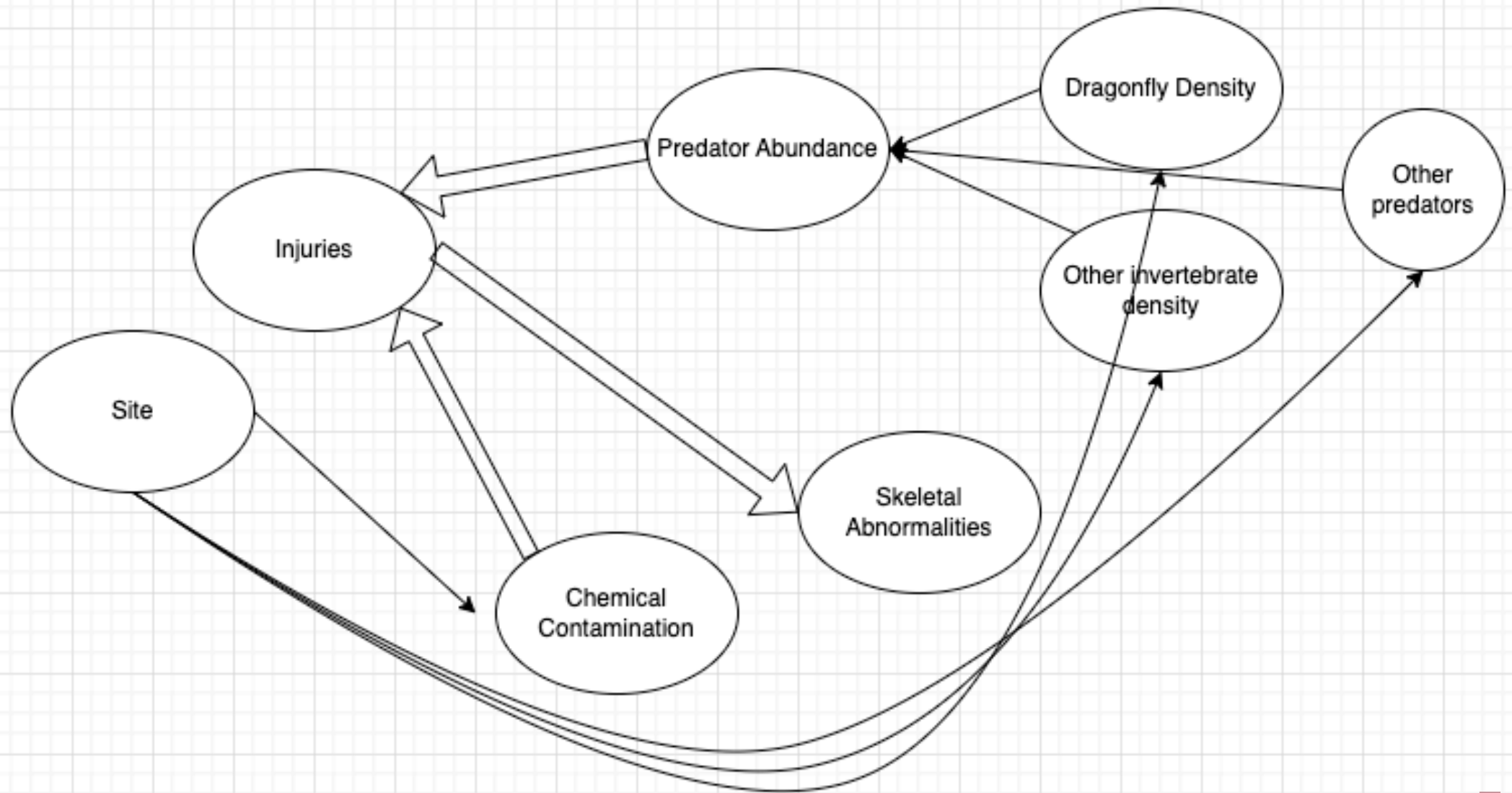
Dataset Size: $n/15$ revised

- To estimate $P(\text{success})$, is ~ 15 rows of data enough? It depends...
- We will definitely need a bigger dataset to estimate the probability of "success" *when success is very rare*
- Let s be the total number of successes in the dataset, and f the number of failures.
- Limiting sample size m is $\min(s, f)$
- Number of coefficients we can estimate is about $\frac{m}{15}$

Model Plan?

The repeated occurrence of abnormal amphibians in nature points to ecological imbalance, yet identifying causes of these abnormalities has proved complex. Multiple studies have linked amphibian abnormalities to chemically contaminated areas, but inference about causal mechanisms is lacking. Here we use a high incidence of abnormalities in Alaskan wood frogs to strengthen inference about the mechanism for these abnormalities. We suggest that limb abnormalities are caused by a combination of multiple stressors. Specifically, toxicants lead to increased predation, resulting in more injuries to developing limbs and subsequent developmental malformations. We evaluated a variety of putative causes of frog abnormalities at 21 wetlands on the Kenai National Wildlife Refuge, south-central Alaska, USA, between 2004 and 2006. Variables investigated were organic and inorganic contaminants, parasite infection, abundance of predatory invertebrates, UVB, and temperature. Logistic regression and model comparison using the Akaike information criterion (AIC) identified dragonflies and both organic and inorganic contaminants as predictors of the frequency of skeletal abnormalities. We suggest that both predators and contaminants alter ecosystem dynamics to increase the frequency of amphibian abnormalities in contaminated habitat. Future experiments should test the causal mechanisms by which toxicants and predators may interact to cause amphibian limb abnormalities. - Reeves et al. 2010, <https://doi.org/10.1890/09-0879.1>

Causal Diagram?



Critique?

```
skeletal_abnormality ~ detectable_analytes +  
  dragonfly_relative_density +  
  other_invert_relative_density
```

- Why not include injuries?
- Would you include site?
- Why isn't the frog's developmental stage in there?

Regression Evolution

old: linear regression

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \beta_3 x_{3,i} + \dots \beta_k x_{k,i} + \epsilon_i$$

- where x s are the k predictor variables,
- β s are the parameters to be estimated by the model,
- and $\epsilon \sim N(0, \sigma)$ are the model residuals.

Regression Evolution

When our response variable was a *count* variable, we modified our equation to:

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \beta_3 x_{3,i} + \dots \beta_k x_{k,i} + \epsilon_i$$

positing that $y_i \sim \text{Pois}(\lambda_i)$ for Poisson regression; similarly for negative binomial regression, we just replaced that Poisson distribution with a negative binomial distribution (and replaced λ_i with μ_i , stating that $y_i \sim \text{NegBin}(\mu_i, \sigma)$).

Binary Response?

What if our response variable is *binary* -- a categorical variable with just two possible values? We will designate one of the two values a "success," and then we want to predict the probability of success as a function of some set of predictors. What will our model equation look like in this case?

Which Distribution?

The Binomial!

$$P(x = s | n, p) = \binom{n}{s} p^s (1 - p)^{(n-s)}$$

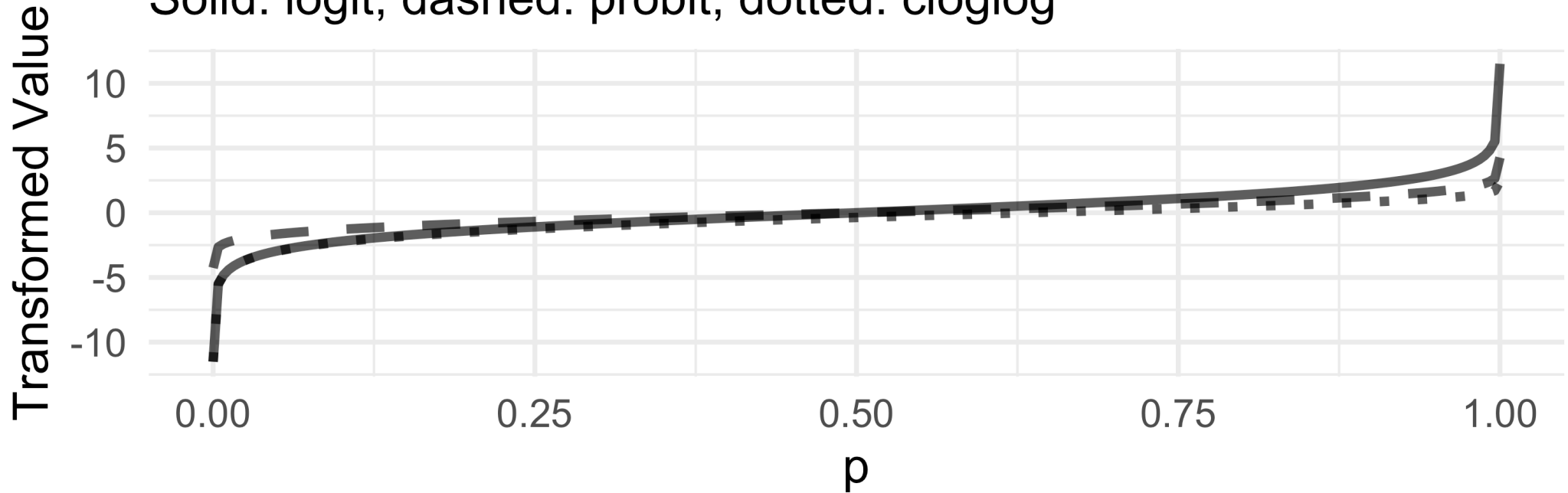
$$X \sim \text{Binom}(n, p)$$

Link Functions?

logit is a common default (depends on discipline)

Link Functions

Solid: logit, dashed: probit, dotted: cloglog



Binary Regression Equation

$$\text{logit}(p_i) = \log\left(\frac{p_i}{(1 - p_i)}\right) =$$

$$\beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \beta_3 x_{3,i} + \dots + \beta_k x_{k,i} + \epsilon_i$$

Where: response $\sim \text{Binom}(n, p_i)$

Back to Frogs...

How does this equation relate back to our desired response variable?

- y_i , the i th observation of the response variable is assumed to follow a binomial distribution with probability p_i
- In other words: $y_i \sim \text{Binom}(n_i, p_i)$
- n_i depends on the setup of the data -- often $n = 1$ for each row of the dataset, as here where each row is one frog. We can think of each frog as one binomial trial, with success/failure meaning abnormality/normality of the frog.

Checking the data setup

- We would like to model the proportion frogs with abnormalities as a function of a set of predictors.
- The variable `skeletal_abnormality` has values "Yes" and "No".
- In R, if we use this (categorical) variable as our response, how will R determine which level (value of the variable) is a "success"?

Response Variable

```
frogs |>  
  # pull out just the variable in question  
  pull(skeletal_abnormality) |>  
  # print out the variable's unique values  
  levels()
```

```
## NULL
```

Binary models *require* a factor() Response Variable (or one that is coded 0, 1)

```
frogs <- frogs |>
  # force our response to be "factor" not "character".
  # This will also auto-sort the levels in to ALPHA ORDER.
  # So DON'T do it AFTER you have carefully re-ordered them!
  mutate(skeletal_abnormality = factor(skeletal_abnormality))
```

```
# to check the results:
frogs |>
  # pull out just the variable in question
  pull(skeletal_abnormality) |>
  # print out the variable's unique values
  levels()
```

```
## [1] "No"  "Yes"
```

Data Prep for Binary Model

Choosing the Order of factor Variable Values

```
# list the values in the "new" order you want  
frogs <- frogs |>  
  mutate(skeletal_abnormality =  
    fct_relevel(skeletal_abnormality, 'No', 'Yes'))
```

```
# if you want to check the result  
frogs |> pull(skeletal_abnormality) |> levels()
```

```
## [1] "No"  "Yes"
```

Model Fitting

Before this, make sure response is a factor

```
frog_model <- glmmTMB(skeletal_abnormality ~  
                      detectable_analytes +  
                      dragonfly_relative_density +  
                      other_invert_relative_density,  
                      data = frogs,  
                      family = binomial(link = 'logit'))
```

```
summary(frog_model)
```

```
## Family: binomial ( logit )
## Formula:
## skeletal_abnormality ~ detectable_analytes + dragonfly_relative_density +
##      other_invert_relative_density
## Data: frogs
##
##           AIC          BIC    logLik -2*log(L)  df.resid
##      1382.8      1406.8    -687.4    1374.8      2918
##
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.802e+00  1.599e-01 -17.515  < 2e-16 ***
## detectable_analytes    -3.270e-03  6.075e-03  -0.538  0.59034
## dragonfly_relative_density    5.209e-03  1.949e-03   2.672  0.00753 **
## other_invert_relative_density -3.357e-05  1.073e-04  -0.313  0.75425
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assessment

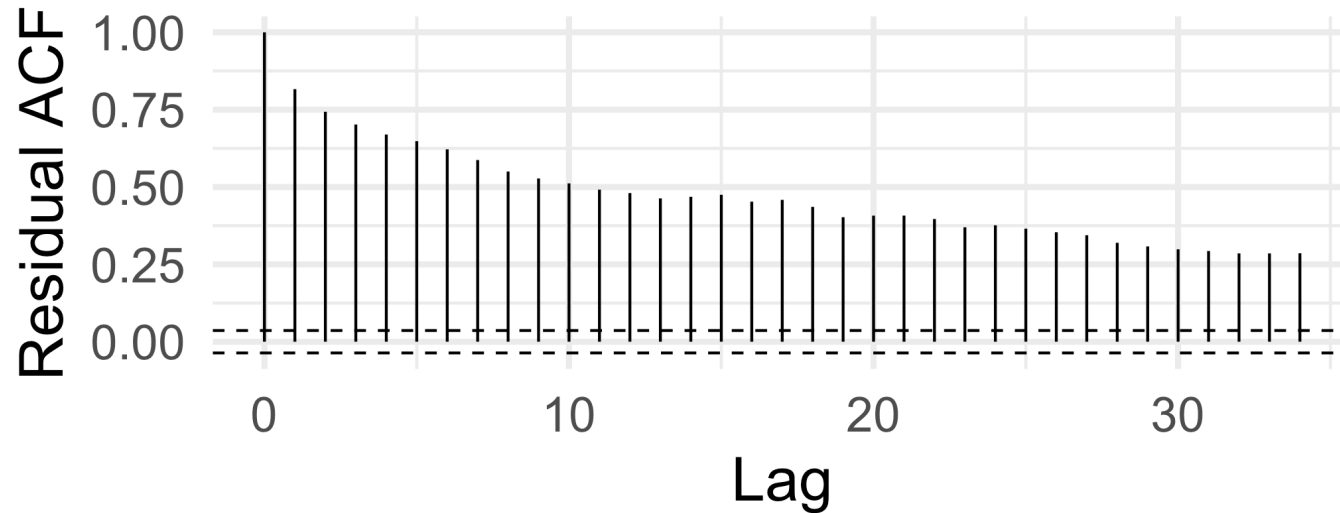
Conditions to Check

- Response is really binary!
- Logit-linearity of predictor, response relationship
- Independence of residuals
- Mean-variance relationship is as expected
 - "mean" is expected response value
 - "variance" is spread of residuals

Assessment

ACF to check independence

```
s245::gf_acf(~resid(frog_model))
```

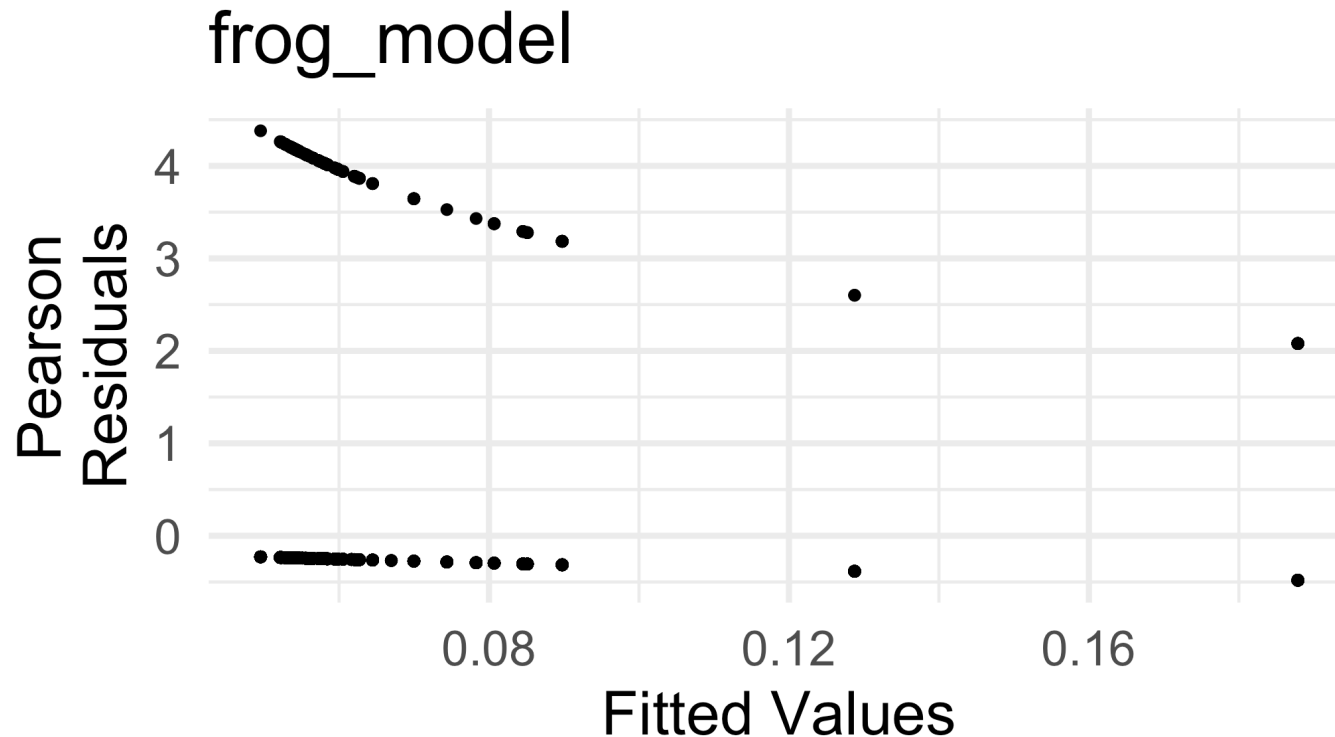


```
# or: acf(resid(frog_model))
```

Do not use this code!

DO NOT USE THIS IN YOUR OWN ASSESSMENT OF YOUR MODELS!

```
gf_point(resid(frog_model, type='pearson') ~ fitted(frog_model)) |>  
  gf_labs(title='frog_model',  
          y=' Pearson\nResiduals', x='Fitted Values')
```



Assessment

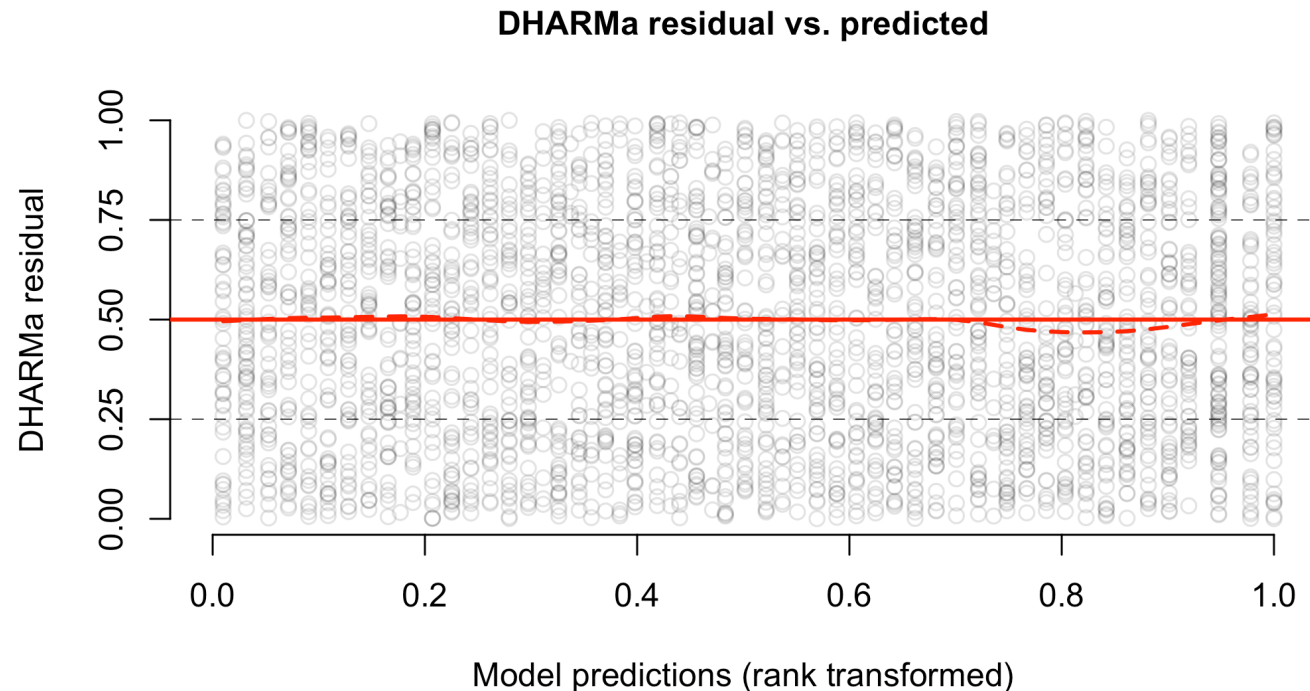
Scaled Residual Plot to check mean-variance, linearity

```
# ACUTALLY USE **THIS** WAY FOR MEAN-VARIANCE CONDITION
```

```
library(DHARMA)
```

```
sim_frog_res <- simulateResiduals(frog_model)
```

```
plotResiduals(sim_frog_res, quantreg = FALSE)
```



Selection

Just as usual.

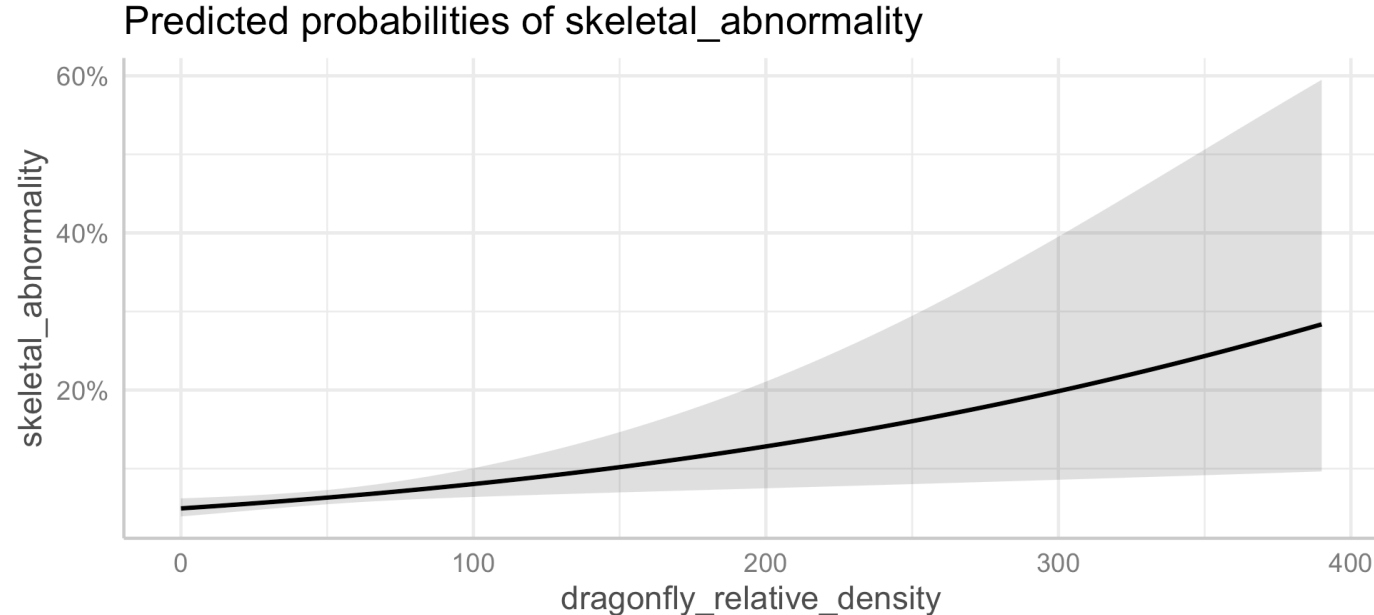
For example, using ANOVA:

```
car::Anova(frog_model)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: skeletal_abnormality
##
##              Chisq Df Pr(>Chisq)
## detectable_analytes      0.2898  1    0.590339
## dragonfly_relative_density  7.1417  1    0.007531 **
## other_invert_relative_density 0.0980  1    0.754254
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

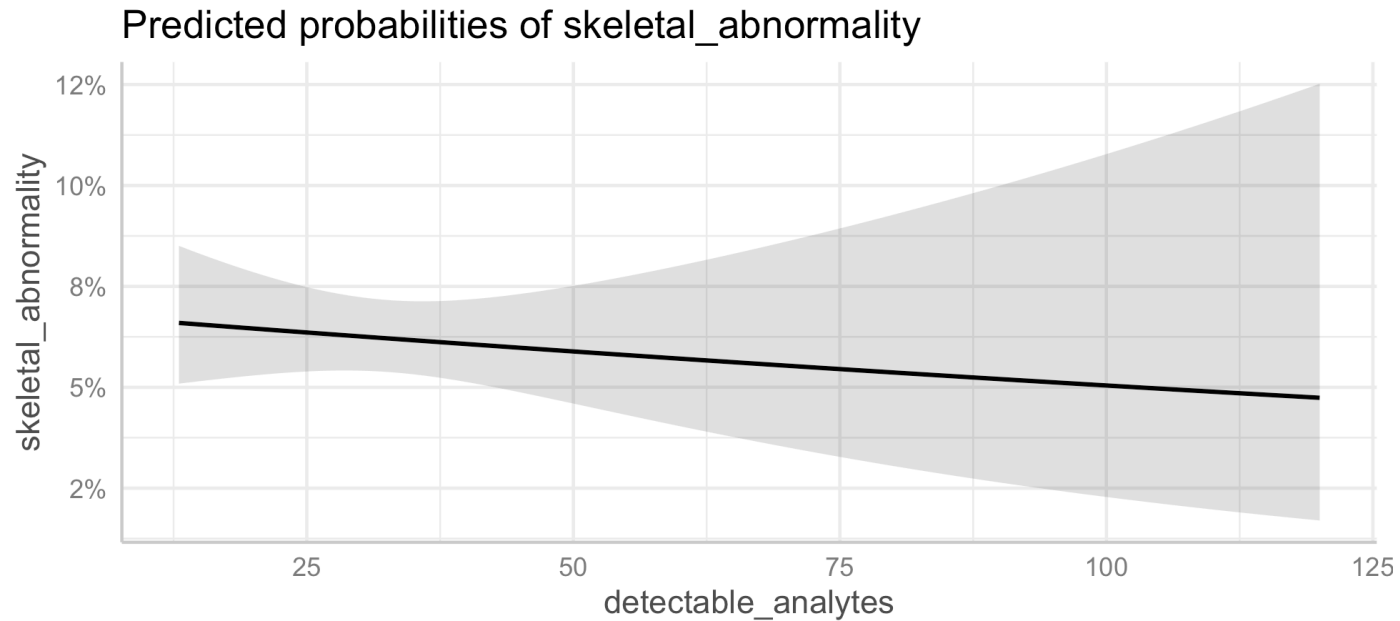
Prediction Plots (same as ever)

```
library(ggeffects)
predict_response(frog_model,
                 terms = 'dragonfly_relative_density') |>
plot()
```



Prediction Plots

```
ggpredict(frog_model,  
          terms = 'detectable_analytes') |>  
plot()
```



Feedback

<https://cutt.ly/arm-feedback>

